

INSE 6450: AI in Systems Engineering

Milestone 2

Due: 11:59 PM (EST), Feb 27, 2026, submit on Moodle.

Include your name and student ID!

- Submit your write-up in PDF (we do not accept hand-written submissions). Text in square brackets are hints that can be ignored.
- All code should be in Python and Pytorch (or Tensorflow).
- Make sure that **all your code for the project** is in a **Github repo** and it **shared with the POD**.

1: Model Selection (2.5 pts)

In Phase 2, you will build on Phase 1 by (a) selecting and justifying an appropriate model, (b) training and evaluating it, and (c) designing how it will be deployed, with clear reasoning about online/offline processing, edge/cloud, batch/streaming, and latency/performance trade-offs. Include performance and efficiency metrics for training and inference, and submit all code used to produce your results.

1. (1 pt) List 2–3 plausible model families for your task (e.g., Logistic, Linear, Tree-based, CNN, RNN, Transformer, GNN, Autoencoder, Recommender, LLM encoder/fine-tune), and answer the following questions:
 - Why it fits your data & objective
 - Why it might not be ideal (e.g., capacity, data size, interpretability, latency, memory)
2. (1 pt) Explain which model you chose and why, citing:
 - Match to task type (classification/regression/forecasting/anomaly/retrieval)
 - How the model's inductive biases fit your features (from Phase 1)
 - Expected accuracy/robustness vs. complexity/latency/memory
3. (0.5 pts) Provide the following details about the architecture:
 - Input representations & preprocessing
 - Main architecture/hyperparameters (e.g., depth, width, embeddings, loss, reg/early stop/aug)
 - Versioned configuration (e.g., seed, learning rate, batch size, optimizer)

2: Training & Experimentation (3.0 pts)

1. (1 pt) Provide the following details about your training procedure:
 - Train/validation/test methodology (random vs. temporal splits; stratification for imbalance)
 - Hyperparameter tuning approach (grid/random/Bayesian/manual)
 - Hardware & environment summary (CPU/GPU, RAM/VRAM), epochs/early stopping criteria
2. (1 pt) Report the following results for your model:
 - Learning curves (loss & relevant metric vs. epoch/steps)
 - At least one ablation/comparison (e.g., baseline vs. final; smaller vs. larger model; key feature on/off)
 - If class imbalance: show mitigation choice (re-weighting, focal loss, sampling) and effect
3. (1 pt) Report appropriate task metrics on validation (and test if fixed):
 - If Classification: accuracy, precision/recall, F1 (macro/micro), AUROC/PR-AUC
 - If Regression/Forecasting: RMSE/MAE/MAPE, R^2

- If Ranking/RecSys/IR: MAP@K, NDCG@K, Recall@K
- Report appropriate results otherwise.

3: Efficiency Metrics (Training & Inference) (2.0 pts)

1. (1 pt) Report the following details regarding training efficiency:
 - Total training time and time/epoch (or steps/sec)
 - Peak memory/VRAM observed
 - Model size (parameters and/or on-disk size)
 - Floating giga point operations (FLOPS)
 - Optional: energy or cost estimate if measurable
2. (1 pt) Measure on your available hardware (specify CPU/GPU & batch size):
 - Latency: p50 and p90 per sample; and batch latency if applicable
 - Throughput: samples/sec under a realistic batch size
 - Memory footprint at inference (RAM/VRAM)
 - Any quantization/distillation/ONNX export attempted—report effect on latency/accuracy

4: Deployment Design & System-Level Trade-offs(2.5 pts)

1. (1 pt) Answer the following questions regarding your model deployment:
 - Online vs. Offline: what runs in real time vs. precomputed
 - Edge vs. Cloud vs. Hybrid: rationale (connectivity, cost, privacy, device limits)
 - Batch vs. Streaming: workload patterns and why your choice fits
2. (0.5 pts) Provide a high-level diagram/description that shows:
 - Data ingestion → preprocessing → model → post-processing → outputs/APIs
 - Where caching, batching, or asynchronous queues are used
 - Failure/rollback plan (e.g., fallback model or heuristic if service degrades)
3. (1 pt) Report the following regarding latency/accuracy trade-offs:
 - Target SLOs (e.g., p95 latency $\leq$ X ms; minimum accuracy/F1 $\geq$ Y)
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 - If applicable: shadow mode or A/B plan (design-only acceptable) and metrics to monitor (drift, error spikes)